



Number Sense Tips & Problem Solving

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Azle ISD – 1974 to 2017

7th grade math, high school math & Physics teacher

Married – 4 children & 3 grandchildren

ETS Physics reader 10 years

Co-founder TMSCA (1981)

Coached all 4 UIL math & science events + slide rule

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UIL Elem & JH Number Sense

Individuals are called upon every day to use their ability to make quick mental calculations to make decisions. The development of such abilities should be an integral part of the math curriculum. Concepts covered include, but are not limited to: addition, subtraction, multiplication, division, proportions, and use of mathematical notation.



UIL EJJ Number Sense

Students will be given a 10-minute, fill-in-the-blank test which they must complete without doing calculations on paper or on a calculator. Erasures and mark-outs are not permitted.

Every tenth problem is an estimation problem with an integral answer and a 5% range of the answer.



Elementary Problem Guidelines

Problem 1 – 20

1. Addition, subtraction, multiplication, & division of whole numbers
2. Recognizing place value
3. Rounding off whole numbers
4. Multiplication short-cuts
5. Remainder type problems
6. Even & odd number type problems
7. Expanded notation
8. Sums of whole numbers with regrouping
9. Differences of whole numbers with regrouping
10. Roman numerals/Arabic numbers conversion



Elementary Problem Guidelines

Problems 21 – 40

1. Addition/subtraction of fractions with common denominators
2. Addition, subtraction, multiplication, & division of decimal fractions
3. Comparing decimal fractions
4. Conversion problems (either way):
fraction/decimal, percent/fraction,
percent/decimal



Elementary Problem Guidelines

Problems 21 – 40 (Continued)

5. Order of operations
6. More multiplication short-cuts
7. Ratio/proportion
8. Consumer type problems
9. Problems about prime numbers
10. Greatest common divisor (GCD) & least common multiple (LCM)
11. Conversion problems (either way): length, measurements, time



Elementary Problem Guidelines

Problems 41 – 60

1. Addition, subtraction, multiplication & division of fractions and mixed numbers
2. Substitution problems
3. Perimeter/area of: square, rectangle, triangle
4. Radius/diameter of a circle
5. Powers & roots of numbers
6. Solving simple equations
7. Sequences
8. Sets
9. Word problems
10. Volume of cube/rectangular box



Elementary Problem Guidelines

Problems 41 – 60 (Continued)

- 11. Right triangle problems
- 12. More multiplication short-cuts
- 13. Base systems conversions



Elementary Problem Guidelines

Problems 61 – 80

1. Addition, subtraction, multiplication & division of integers
2. Inverses
3. Basic geometry facts
4. More area problems
5. Squaring two-digit numbers
6. More multiplication short-cuts
7. More powers and roots of numbers
8. More consumer type problems
9. Inequalities



Elementary Problem Guidelines

Problems 61 – 80 (Continued)

- 10. Probability
- 11. More area problems: parallelogram, rhombus, trapezoid
- 12. Coordinate geometry - number line
- 13. More percent type problems



Junior High Problem Guidelines

Problems 1 – 20

1. Addition, subtraction, multiplication & division of whole numbers, fractions, and decimals
2. Order of operations
3. Use of the distributive property
4. Comparison of fractions & decimals
5. Multiplication short-cuts
6. Squaring numbers
7. Roman numerals/Arabic numbers
8. Mean, median, mode
9. Sums of whole numbers of multiple terms of a finite series



Junior High Problem Guidelines

Problems 21 – 40

1. Addition, subtraction, multiplication & division of mixed numbers and integers
2. More multiplication short-cuts
3. Percent problems
4. Conversion problems (either way):
English/metric, length, area, capacity, time
5. Consumer type problems
6. Substitution problems
7. Solving simple equations



Junior High Problem Guidelines

Problems 21 – 40 (Continued)

- 8. Square roots/cube roots
- 9. Greatest common divisor (GCD) & least common multiple (LCM)
- 10. Number theory - prime numbers and divisors
- 11. Perimeter/area of: square, rectangle, circle
- 12. Ratio/proportion
- 13. Inverses
- 14. Multiplication of 101, 111



Junior High Problem Guidelines

Problems 41 – 60

1. Sets
2. Word problems
3. Pythagorean theorem
4. Sequences
5. Volume/surface area of rectangular solid/cube
6. Base systems: conversions and basic operations
7. Area of: parallelogram, rhombus, trapezoid, circle



Junior High Problem Guidelines

Problems 41 – 60 (Continued)

- 8. Solving inequalities
- 9. Basic geometry facts
- 10. Remainder problems



Junior High Problem Guidelines

Problems 61 – 80

1. Repeating decimals
2. More number theory
3. Powers of numbers
4. Volume of: circular cylinder, cone, sphere
5. Sequences & series
6. Factorial
7. Coordinate geometry
8. Probability/odds
9. More percent type problems : Advanced
10. More remainder type problems
11. More multiplication short-cuts

Problems

(1) $25 \times 8 =$ _____

$$8 \div 4 = 2$$

$$2 \times 100 =$$

200

Problems

(2) $75 \times 23 =$ _____

$$23 \div 4 = 5.75$$

$$5.75 \times 300$$

$$75 = \frac{300}{4}$$

1725

Problems

(3) XXIX = _____ Arabic Number

M = 1000; D = 500; C = 100; L = 50;
X = 10; V = 5; I = 1

XXIX = 20 + 9

= 29

Problems

(4) $\frac{11}{8} - \frac{5}{8} =$ _____(fraction)

$$\frac{6}{8}$$

$$\frac{6}{8} \div \frac{2}{2}$$

$$\boxed{\frac{3}{4}}$$

Problems

(5) $24 \times 26 =$ _____

Since $26 = 25 + 1$ and

$$24 = 25 - 1$$

$$24 \times 26 = (25 + 1)(25 - 1)$$

$$24 \times 26 = 25^2 - 1^2$$

$$24 \times 26 = 625 - 1$$

624

Problems

(6) $24 \times 26 = \underline{\hspace{2cm}}$ (Another Way) $\underline{\hspace{2cm}}$

Since ten's digits are the same

And one's digits add up to 10

$4 \times 6 = 24$ – write this down $\underline{\hspace{1cm}}24$

$2 \times (2 + 1) = 6$ – write this down for
the finished answer

624

Problems

(7) $24 \times 26 = \underline{\hspace{2cm}}$ (Still Another Way) $\underline{\hspace{2cm}}$

$4 \times 6 = 24$ – write down 4 and
(*LAST*) keep 2 in your memory $\underline{\hspace{1cm}}4$

$(4 \times 2) + (2 \times 6) + 2 = 22$ – write down
(*INNER + OUTER*) 2 and keep $\underline{\hspace{1cm}}24$
2 in your
memory

$(2 \times 2) + 2 = 6$ – write this down for
(*FIRST*) the finished answer

624

Problems

(8) $.121212\dots = \underline{\hspace{2cm}}$ (fraction)

$$.121212\dots = \frac{12}{99}$$

$$\frac{12}{99} \div \frac{3}{3}$$

$$\frac{4}{33}$$

Problems

$$(9) \quad 1 + 2 + 3 + \dots + 9 = \underline{\hspace{2cm}}$$

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$\text{SUM} = \frac{9(9+1)}{2} \quad \mathbf{45}$$

Note: You should know the formulas for sums of odd and even integers also !

Problems

(10) $14 + 17 + 20 + 23 + 26 =$ _____

For sums of equally spaced numbers, multiply the **median** of the numbers by the number of terms.

$$\text{SUM} = 20 \times 5$$

100

Problems

(11) $21 + 24 + 27 + 30 =$ _____

For sums of equally spaced numbers, multiply the **median** of the numbers by the number of terms.

$$\text{SUM} = 25\frac{1}{2} \times 4$$

102

Problems

(12) $4\frac{1}{3} \times 4\frac{2}{3} = \underline{\hspace{2cm}}$ (mixed number)

$$\frac{1}{3} \times \frac{2}{3} = \frac{2}{9}$$

Write this down

$$20\frac{2}{9}$$

$$4 \times (4 + 1) = 20 \text{ Write this down}$$

$$20\frac{2}{9}$$

Problems

(13) $6\frac{1}{3} \times 3\frac{1}{3} = \underline{\hspace{2cm}}$ (mixed number)

$$\frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$$

Write this down

$$21\frac{1}{9}$$

$$6 \times 3 + (6 + 3)\frac{1}{3} = 21 - \text{Write this down}$$

$$21\frac{1}{9}$$

Problems

(14) 16% of 36 is 8% of _____

In equation form
looks like this:

$$16\% \times 36 = 8\% \times ?$$

Solving for ? $\rightarrow$ $\frac{16\%}{8\%} \times 36 = ?$

$$\frac{2}{1} \times 36 = ?$$

72

Problems

(15) 5 base 8 + 7 base 8 = _____ base 8

$$5 + 7 = 12$$

$$12 \div 8 = 1 \text{ remainder } 4$$

Write down 4 and “carry” 1

14

14

Problems

(16) The radius of a circle with an area of 16π is_____

$$A = \pi r^2$$

$$r = \sqrt{\frac{16\pi}{\pi}}$$

4

Problems

(17) $12 \div 4 \times 3 =$ _____

$$12 \div 4 = 3$$

$$3 \times 3$$

9

(18) $12 + 4^2 \times 3 =$ _____

$$4^2 = 16$$

$$16 \times 3 = 48$$

$$12 + 48$$

60

Problems

(19) $26 \times 86 =$ _____

Since one's digits are the same

And ten's digits add up to 10

$6 \times 6 = 36$ – write this down ____36

$(2 \times 8) + 6 = 22$ – write this down for
the finished answer

2236

Problems

(20) $113^2 \div 4$ has a remainder of _____

$$13 \div 4 \rightarrow \text{remainder} = 1$$

$$1^2 \div 4 = 0 + \text{remainder}$$

1

Problems

(21) $(13^2 + 11 \times 15) \div 7$ has a remainder of _____

$$13 \div 7 \rightarrow \text{remainder} = 6$$

$$6^2 \div 7 \rightarrow \text{remainder} = 1 \quad (1 + 4 \times 1) \div 7$$

$$11 \div 7 \rightarrow \text{remainder} = 4$$

$$15 \div 7 \rightarrow \text{remainder} = 1 \quad = 0 + \text{remainder}$$

5

Problems

(22) How many total subsets can be made of $\{A, U, S, T, I, N\}$? _____

The set has 6 elements, so the number of subsets is

$$2^6$$

64

Problems

(23) The area of a rhombus with diagonals 17 and 20 is _____

$$\text{Area of a rhombus} = \frac{\text{diagonal (1)} \times \text{diagonal (2)}}{2}$$

$$A_{\text{rhombus}} = \frac{17 \times 20}{2}$$

64

Problems

(24) What is the area of a square with diagonal 14? _____

$$\text{Area} = \frac{(\text{diagonal})^2}{2}$$

$$A = \frac{14^2}{2} \quad \mathbf{98}$$

Problems

(25) What is the length of the side of an equilateral triangle with area $9\sqrt{3}$?

$$\text{Area} = \frac{(\text{side})^2 \sqrt{3}}{4}$$

$$\text{side} = \sqrt{\frac{4 \text{ Area}}{\sqrt{3}}}$$

$$\text{side} = \sqrt{\frac{4 \cdot 9\sqrt{3}}{\sqrt{3}}}$$

6

Problems

(26) $91 \times 96 =$ _____

$100 - 91 = 9$ and $100 - 96 = 4$.

Multiply 9×4 and write down. 
8736

Subtract 9 from 96 or 4 from 91 and write down. 

8736

Problems

(27) $6 \frac{3}{4} \div \frac{1}{4} = \underline{\hspace{2cm}}$

Recall $\div \frac{1}{4}$ is same as multiplying by 4

Also recall $6 \frac{3}{4}$ is the same as $(6 + \frac{3}{4})$

$$(6 + \frac{3}{4}) \times 4 = 24 + 3 \quad \text{27}$$

Problems

(28) $\frac{5}{9} + \frac{9}{5} = \underline{\hspace{2cm}}$ (mixed number)

Write down the number 2 for the whole number part of the answer

$$2 \frac{16}{45}$$

$$2 \frac{16}{45}$$

Square the difference between the numerator and denominator and place this over the product of the numerator and denominator as the fraction part of the answer.

Problems

(29) 100101110 base 2 = _____base 8

Starting at the right end of the number group the digits into sets of 3 digits.

100	101	110
4	5	6

Convert each of the sets from base 2 to base 8 numbers and write down as final answer.

456

Estimation Problems

*(30) $69 + 79 + 199 =$ _____

$\begin{array}{ccc} \text{---} & \text{---} & \text{---} \\ \downarrow & \downarrow & \downarrow \\ 70 & + & 80 & + & 200 \end{array}$

350

330 – 364

Estimation Problems

*(31) $624 \times 240 =$ _____

Recall $\frac{5}{8} = .625$

$$\frac{5000}{8} \times 240 = 150\,000$$

142 272 – 157 248

Estimation Problems

*(32) $101^2 - 99^2 =$ _____

$$(101 - 99) \times (101 + 99)$$

$$(2) \times (200)$$

$$= 400$$

$$380 - 420$$

Estimation Problems

*(33) $167 \times 359 + 33 =$ _____

$$\frac{1}{6} \approx .167$$

$$\frac{1000}{6} \times 360 + 0$$

$$56\,987 - 62\,985$$

$$= 60\,000$$

Estimation Problems

$$*(34) \quad 269 \times 3\frac{5}{9} = \underline{\hspace{10em}}$$

$$270 \times \frac{32}{9}$$

$$270 \div 9 = 30$$

$$30 \times 32 = 960$$

909 – 1005

Estimation Problems

*(35) $\sqrt{224} \times \sqrt{325} =$ _____

Recall: $15^2 = 225$ and $18^2 = 324$

$$15 \times 18 = 270$$

257 – 283

Estimation Problems

$$*(36) \quad 83 \frac{1}{3} \times 2390 = \underline{\hspace{2cm}}$$

$$\text{Recall: } \frac{5}{6} = .83333 \dots$$

$$\frac{500}{6} \times 2400 = 200000$$

$$\mathbf{189209 - 209125}$$

Practice Problems

(1) $25 \times 32 =$ _____

(2) $1 + 2 + 3 + \dots + 19 =$ _____

(3) $97 \times 93 =$ _____

(4) What is the area of a square with diagonal 8? _____

(5) $DCLX + IX =$ _____ (Arabic Number)

(6) $17 \times 97 =$ _____

(7) $113^2 \div 9$ has a remainder of _____

(8) $77 \times 73 =$ _____

*(9) $119 \times 165 =$ _____

*(10) $119 \times 251 =$ _____

800; 190; 9021; 32; 669; 1649; 7; 5621; 18654-20616; 28376-31362



Some Resources

AMT Test Writing Service

- 675 Miller Rd., Azle, TX 76020
- entermeet@gmail.com
- Phone: 817-444-3655

Offers Number Sense: *Elements of Number Sense* by Jim Cummings. Contains preparatory material for the Number Sense Contest



Some Resources

D & R Enterprises

- 1101 W. Monte Cristo Rd. West, Edinburg, TX 78541
- Phone: 956-383-0372

No Sense in Mathematics (4th edition). By Don Skow



Some Resources

Hexco, Inc.

- PO Box 199, Hunt, TX 78024-0199
- 800/725-2627; Fax: 830-367-3824
- Email: hexco@hexco.com Web site: www.hexco.com

Supplies materials for Dictionary Skills, Number Sense, and Spelling contests. Offers Dictionary Skills and Spelling practice tests. Also available, Spelling Complements for each graded list containing all the dictionary work for the *A+ Spelling List* and for Word Power, plus audio tapes, spelling software and spelling rules book. For Number Sense, offers software and practice tests.



Some Resources

Mental Mathematics for Number Sense

- Frances Walzel
- 2023 CR 08, Cameron, TX 76520
- E-Mail: walzel@vvm.com

Offers 103-page booklet of problems, keys and coded pages for elementary and secondary number sense.



Some Resources

MRC Jr.

- 1024 Scenic Drive, Justin, TX 76247
- Phone: 940-648-8587; Fax: 940-648-8580
- Email: tomcat2243@ev1.net

Offers study materials and tests for Maps, Graphs and Charts, Dictionary Skills, 5/6 and 7/8 Social Studies, Science I, Science II, Number Sense and Mathematics.



Some Resources

Dr. Numsen/Doug Ray

- PO Box 312578, New Braunfels, TX 78131
- Phone: (512) 797-2158; Fax: (208) 575-9617
- Email: doug@academicmeet.com
- Web site: www.academicmeet.com

Provides workbooks and practice tests for elementary and junior high Number Sense, Calculator Applications and Mathematics.



Some Resources

Leo Ramirez

- 9801 W. Parmer Lane #2622, Austin, TX 78717
- Phone: (956) 491-3155 (cell)
- Email: toywiz127@aol.com
- Website: <http://www.rammaterials.com/>

Number Sense, Calculator Applications, Mathematics, and Science Workbooks (including Number Sense: A Starter's Kit, Middle School Magic, Number Sense Magic, Revised Calculator Applications workbook), DVDs and practice tests. Mr. Ramirez is available for writing invitational meet tests and conducting workshops.



Some Resources

TMSCA Test Pool

- Texas Math/Sciences Coaches Association
- PO Box 206, Olney Texas, 76374-0206
- (940) 563-1005
- TMSCA.org

Offers study materials for math, number sense, calculator and science contests.



Some Resources

The handbook, *Developing Middle School Number Sense Skills*, is available. It is the same edition first published in 1996. Stock #217. Cost: \$6.00.

University Interscholastic League

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